

# **CS611 Machine Learning Engineering**

## **Lecture 5: Model Deployment**

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# Outline

Class Case Study #1 - 1 hour

Class Case Study #2 - 45 min

Lecture - 40 mins

- Model Deployment

- Types of Deployments

- Inference pipelines

- Scaling

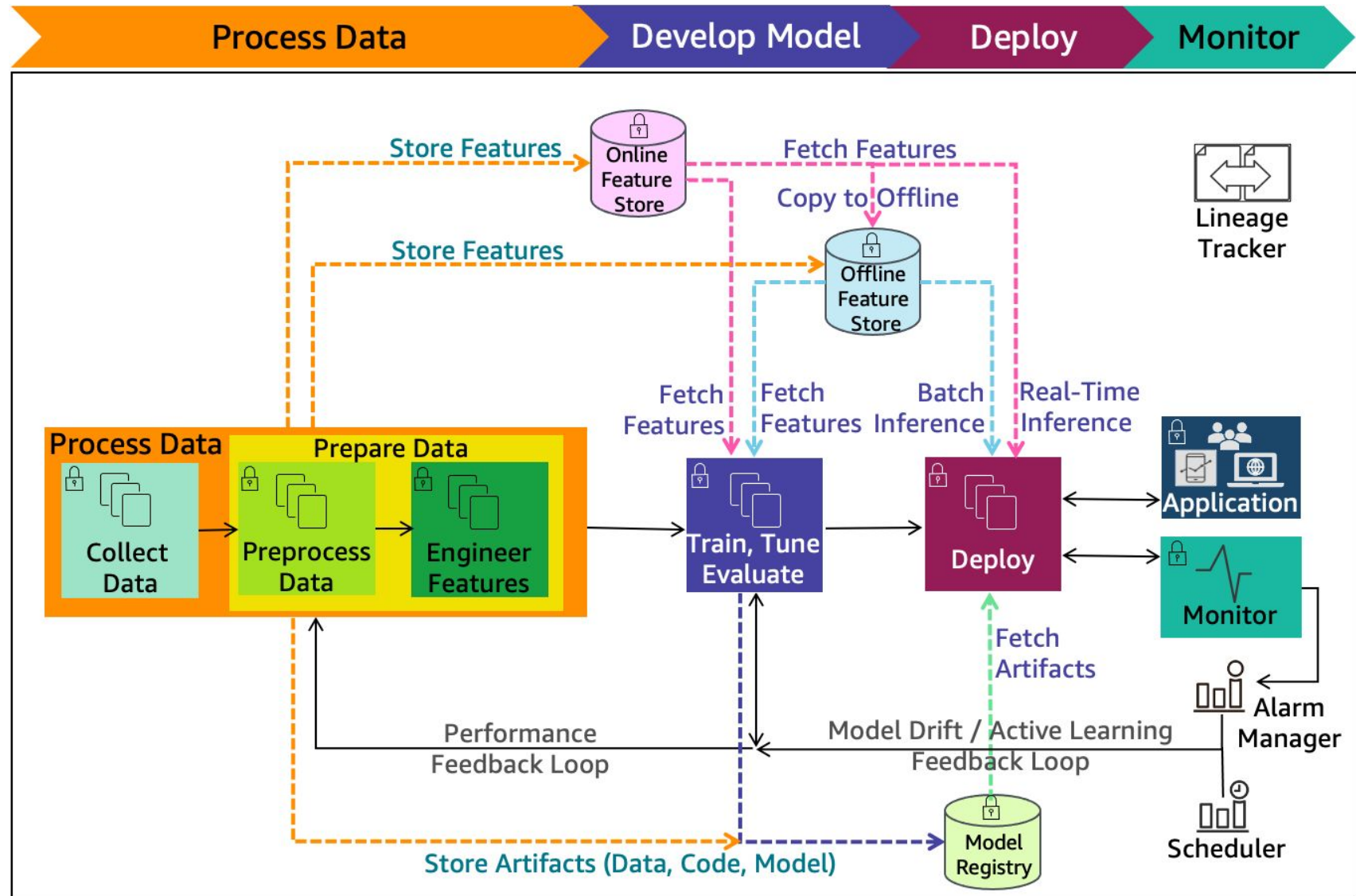
- Deployment Strategies

- Model Registry

- Model Artefacts

Lab - 20 mins

# Machine Learning Lifecycle



# Case Study Context

You are the Data Scientist at a bank.

You are tasked to build a credit scoring model to predict if a customer will default on a loan.

At the moment, you have access to the loan management system which shows daily repayments of loans by customers.

# Model Deployment

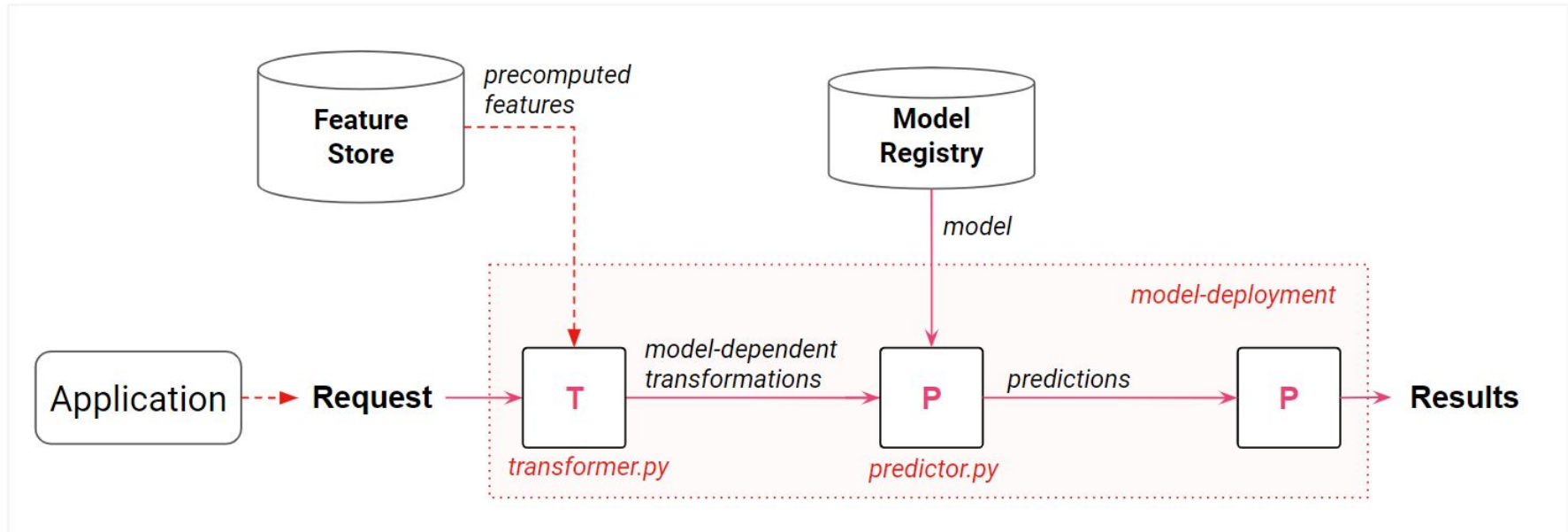
# **Class Discussion:**

## **How do you deploy a batch model?**

# **Class Discussion:**

## **How do you deploy a real-time model?**

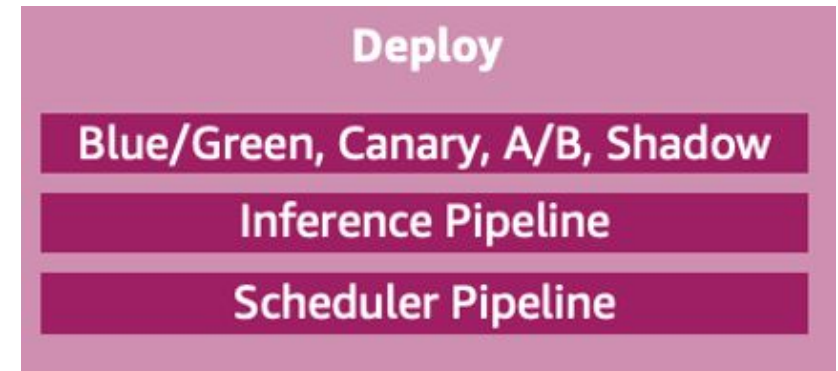
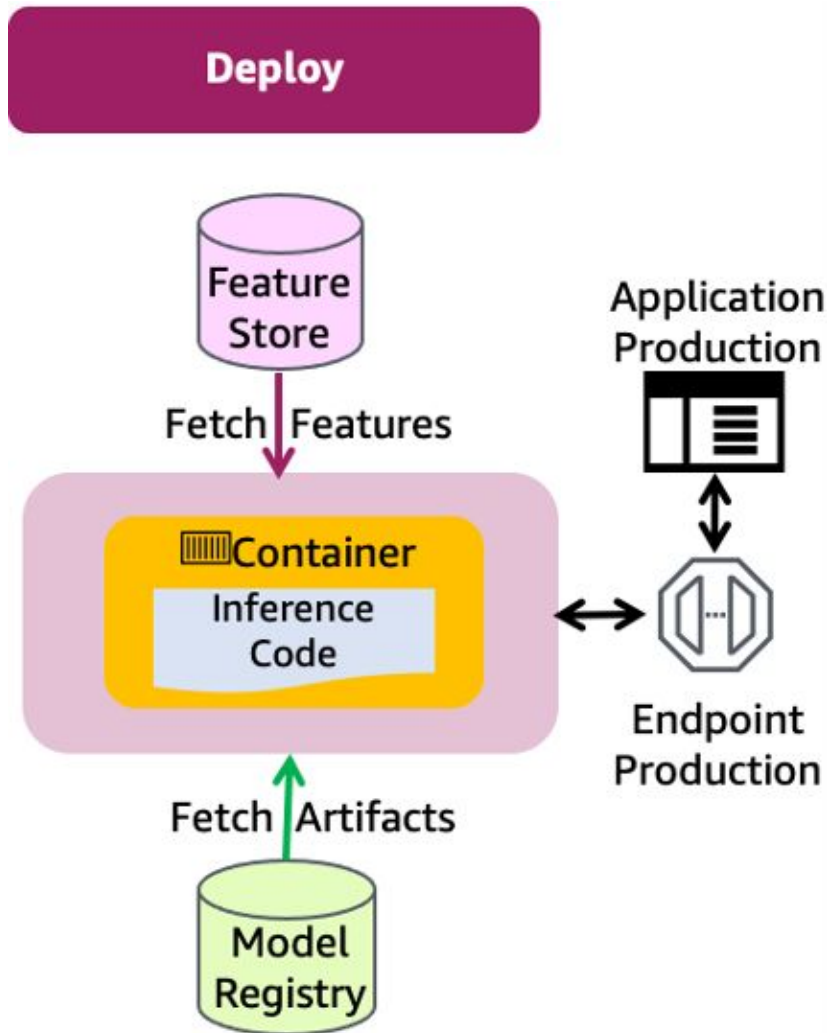
# Model Deployment



**TPP** = Transform the input request into features, Predict using input features and the model, Post-process predictions, before output results.

Source: <https://www.hopsworks.ai/dictionary/model-deployment>

# Model Deployment



# Types of Deployments

# Types of Model Deployments

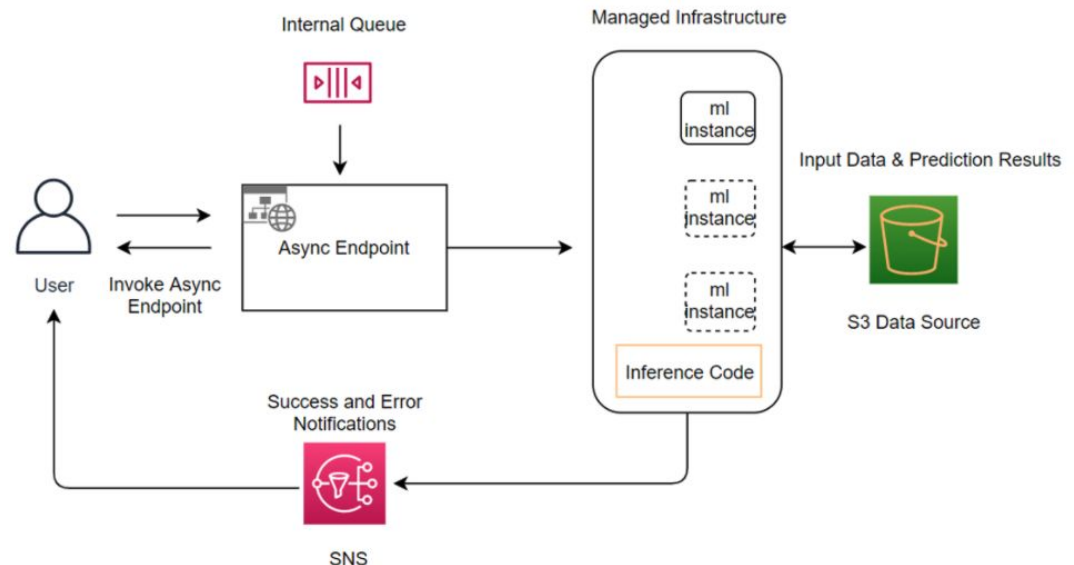
- In general:
  - Batch inference
  - Online inference
    - Real-time inference
    - Asynchronous inference
    - Serverless inference

# Types of Model Deployments

- Batch inference:
  - Running offline on a large dataset.
  - No latency requirements. We can run the inference and wait for the results in few hours.
  - Normally based on a schedule or event-based trigger.
  - We can remove all the infrastructure after the process completed.
- Real-time inference [online]
  - Real-time, interactive, low-latency requirements.
  - The model needs to be available and ready all the time.
  - Predict one data input at a time, but we may have many requests at a time.

# Types of Model Deployments

- Asynchronous inference [online]
  - Typically used in large input requiring longer processing times such as in complex NLP and computer vision use cases.
  - Operate in near real time by using an internal queue.
  - Prediction is stored in Amazon S3 storage, and user gets the results from there.



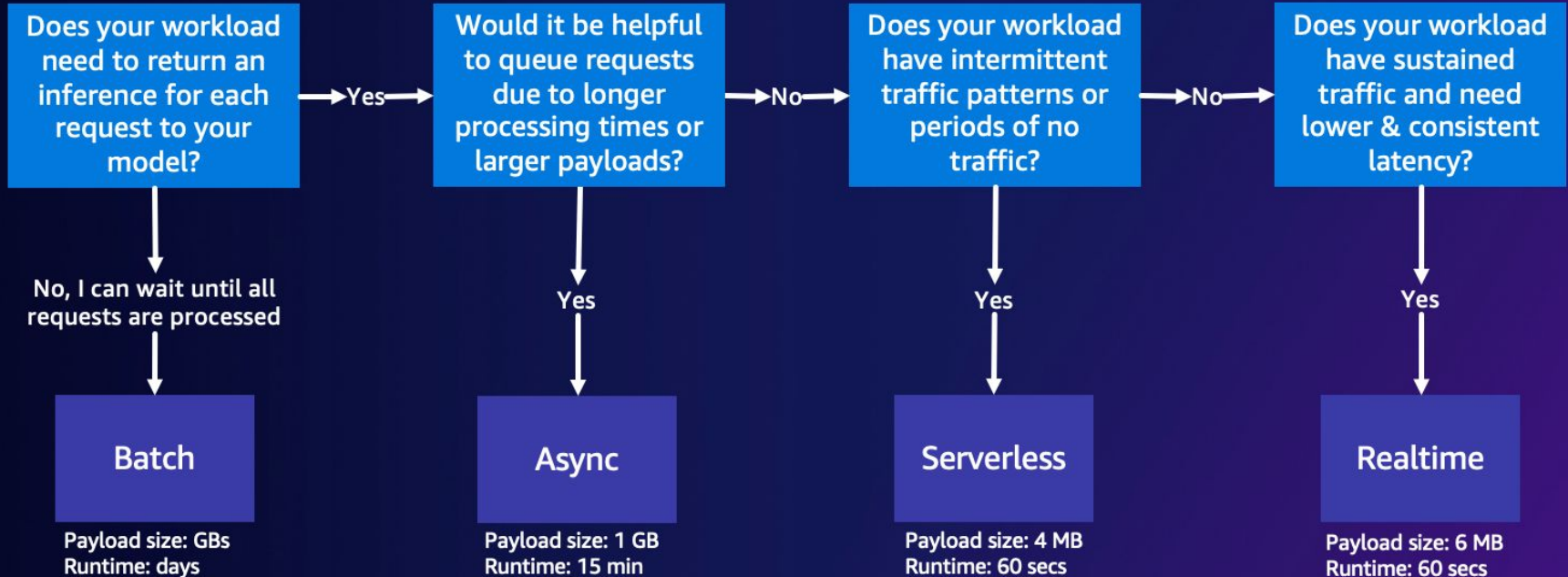
More information: <https://docs.aws.amazon.com/sagemaker/latest/dg/async-inference.html>

# Types of Model Deployments

- Serverless inference [online]
  - Real-time inference that have idle periods between traffic spurt and can tolerate a few extra seconds of latency during first invocation after an idle period (cold start).
  - Automatically scale in and out depending on traffic
  - During an idle time, compute capacity is not running.

# Choosing Model Deployment

## Choosing Model Deployment Options



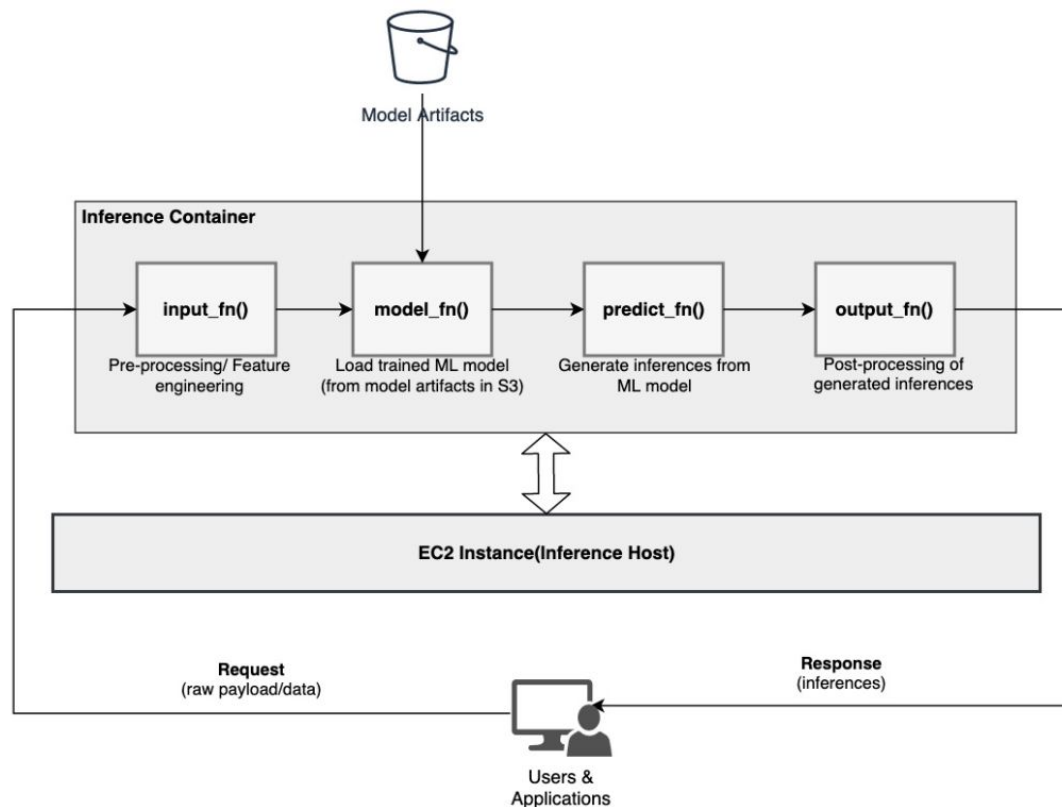
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Source:  
<https://aws.amazon.com/blogs/machine-learning/part-2-model-hosting-patterns-in-amazon-sagemaker-getting-started-with-deploying-real-time-models-on-sagemaker/>

# Inference Pipelines

# Inference Pipeline

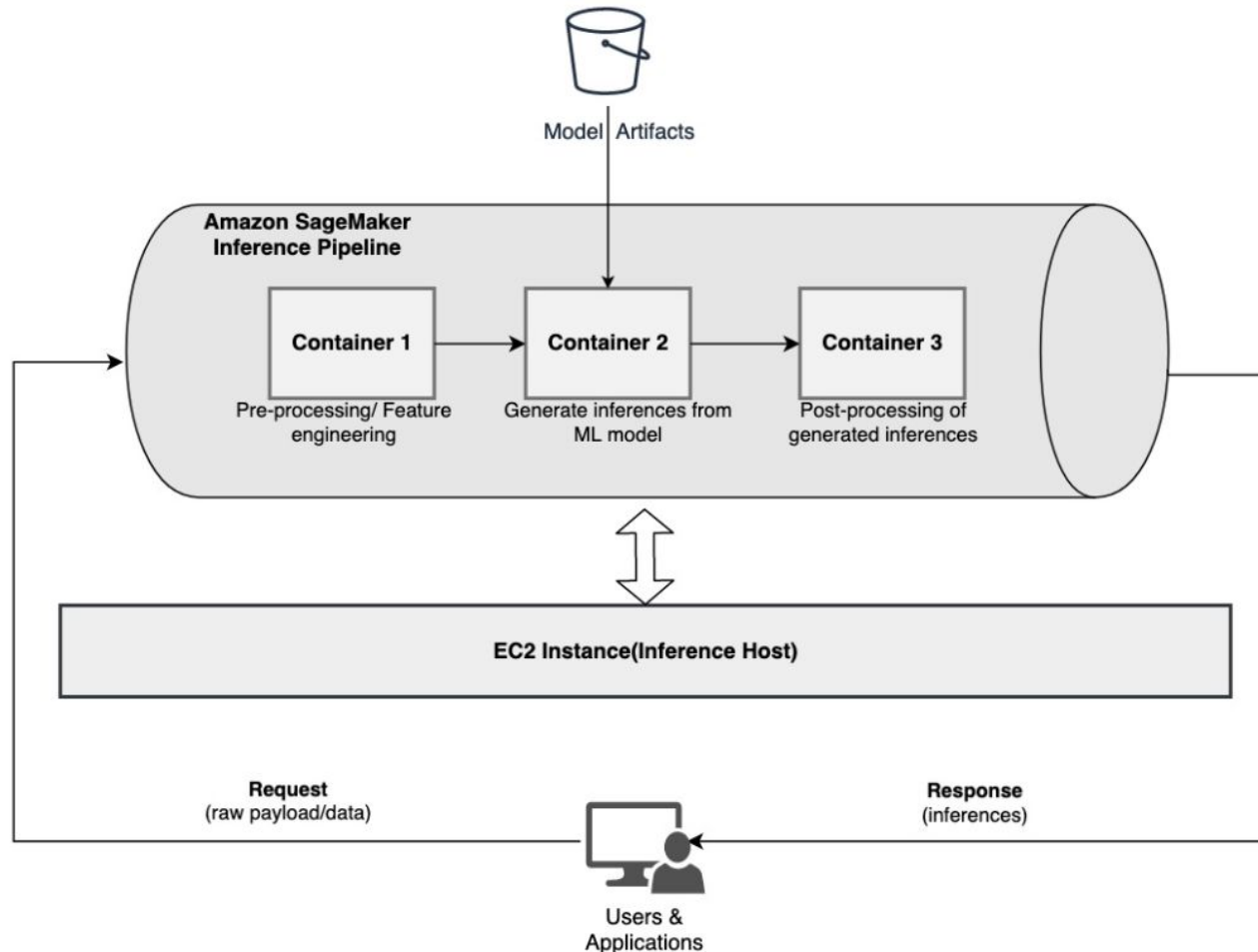
Sometimes we need to perform pre-processing and post-processing after prediction



Source:

<https://aws.amazon.com/blogs/machine-learning/part-4-model-hosting-patterns-in-amazon-sagemaker-design-patterns-for-serial-inference-on-amazon-sagemaker/>

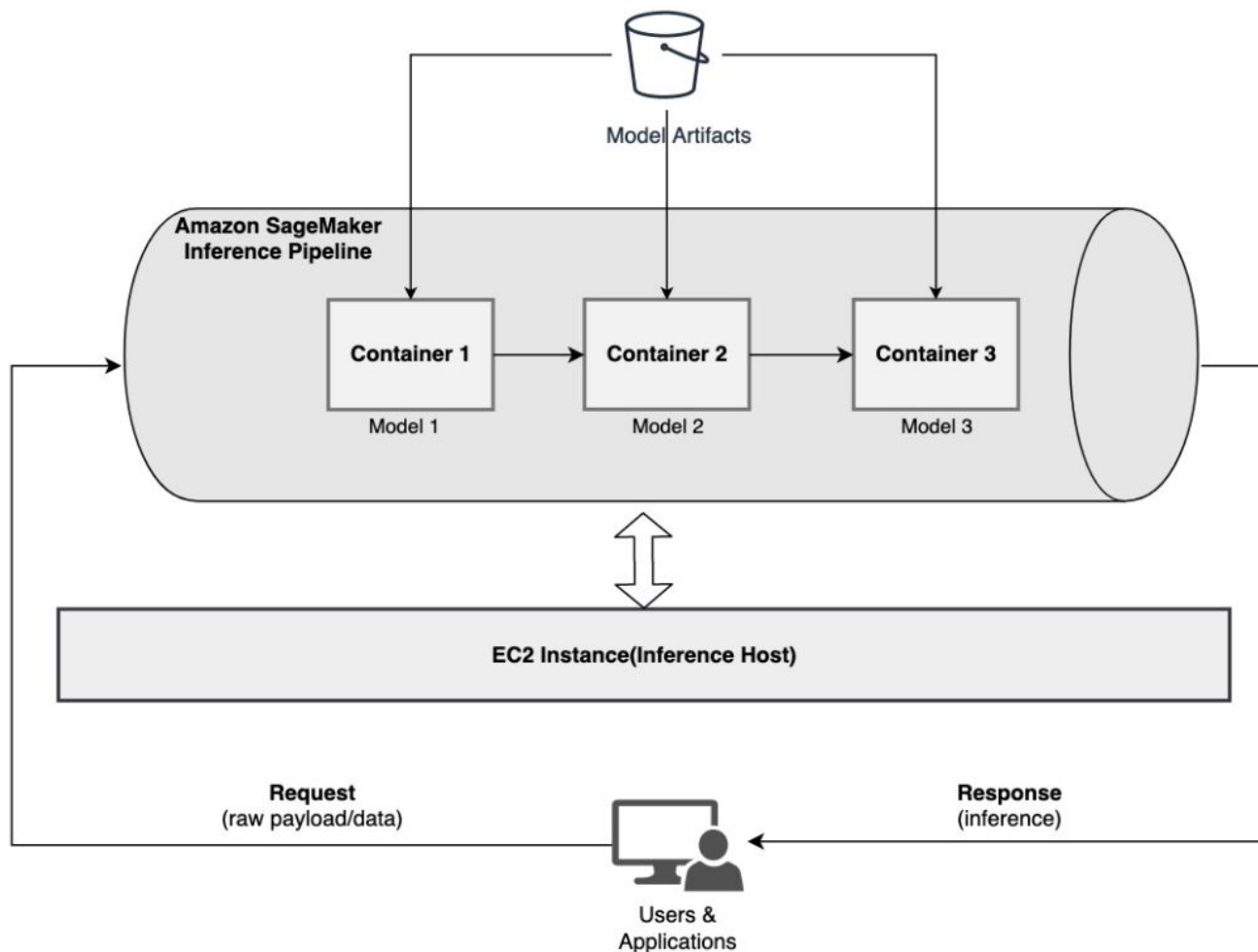
# Inference Pipeline with Multiple Containers



Source:

<https://aws.amazon.com/blogs/machine-learning/part-4-model-hosting-patterns-in-amazon-sagemaker-design-patterns-for-serial-inference-on-amazon-sagemaker/>

# Inference Pipeline for Model Ensemble



Source:

<https://aws.amazon.com/blogs/machine-learning/part-4-model-hosting-patterns-in-amazon-sagemaker-design-patterns-for-serial-inference-on-amazon-sagemaker/>

# Scaling

# Scaling

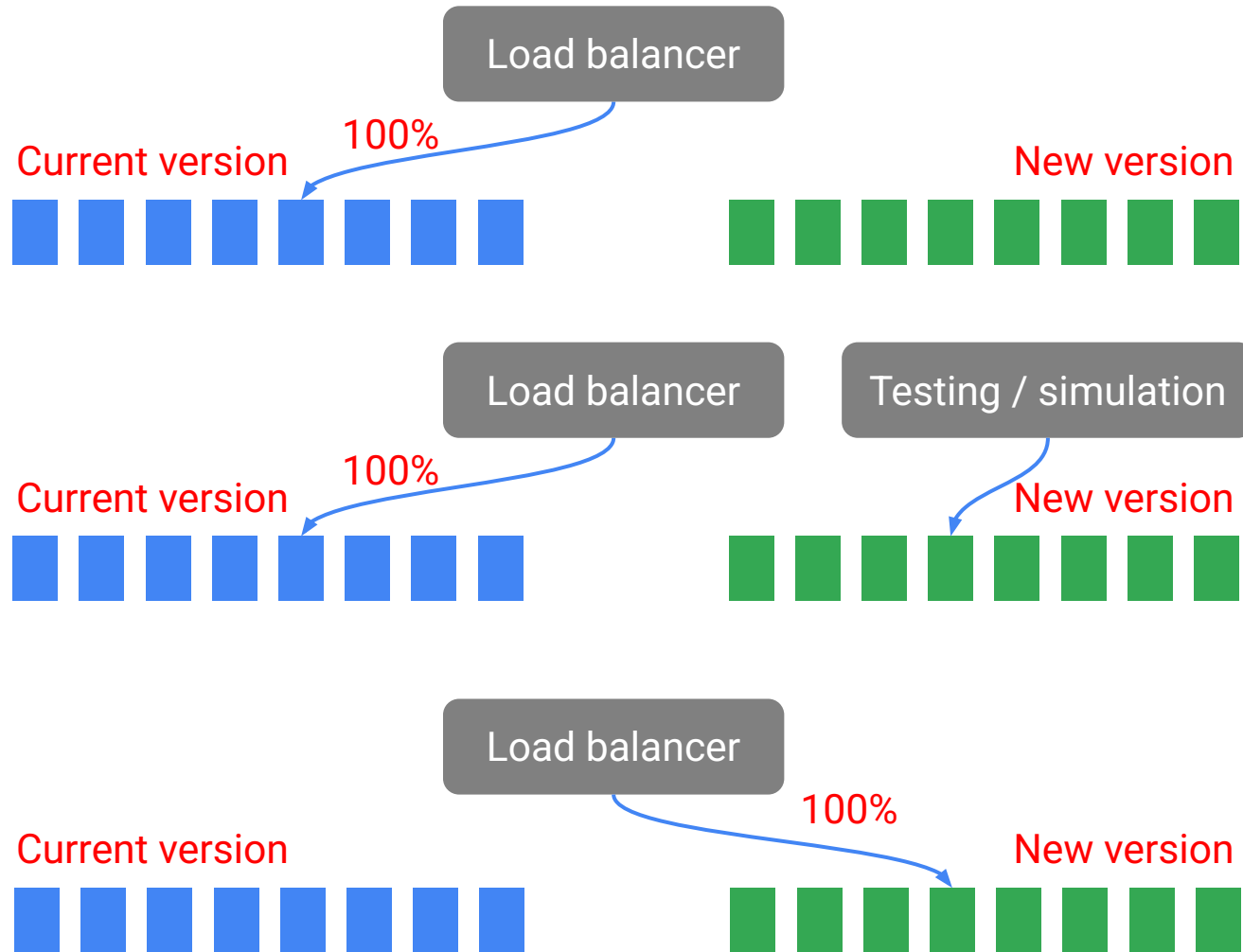
- Just like a web application, real-time inference may expect large number of requests at a time.
- The system needs to be able to scale out, having more machines to meet the demand.
- The system needs to be able to scale in during low period to save costs.
- We can use traffic monitoring to help us to make data driven decision to scale in and out.

# Deployment Strategies

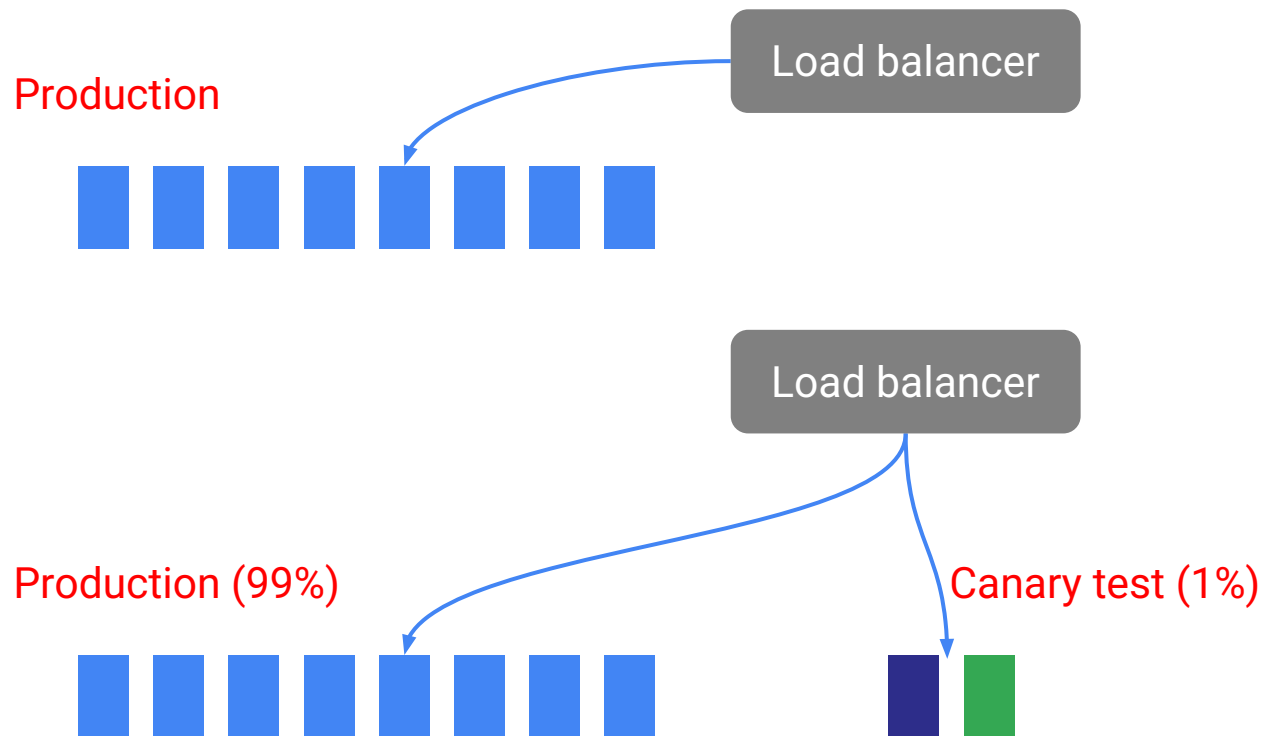
# Model Deployment

- Deployment strategies to reduce downtime and risks when releasing a new or updated version:
  - Blue/green deployment
  - Canary deployment
  - A/B testing
  - Shadow deployment
- Inference pipeline allows us to combine pre-processing and post-processing after machine learning prediction
- Remember that our deployed model needs to be updated from time to time, and automation greatly helps in deployment

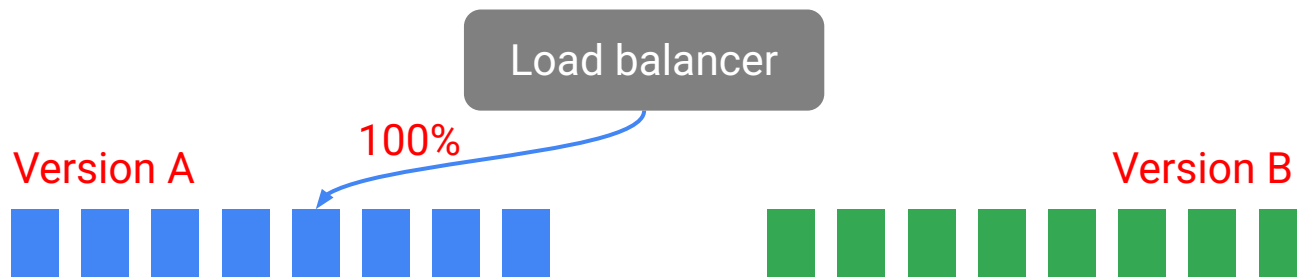
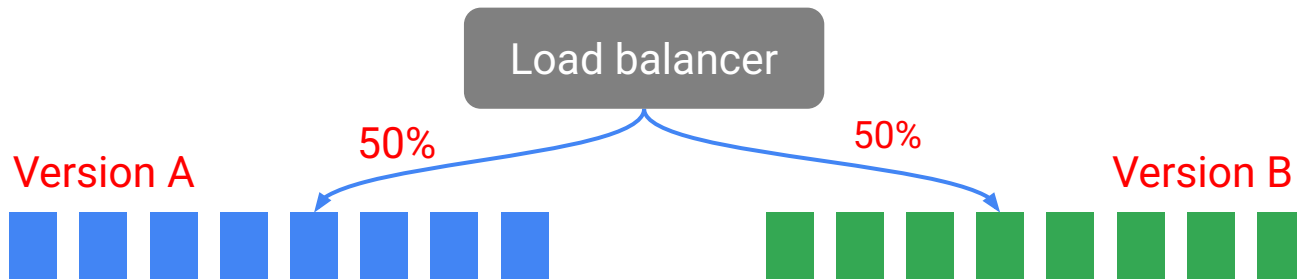
# Blue/green Deployment



# Canary Deployment



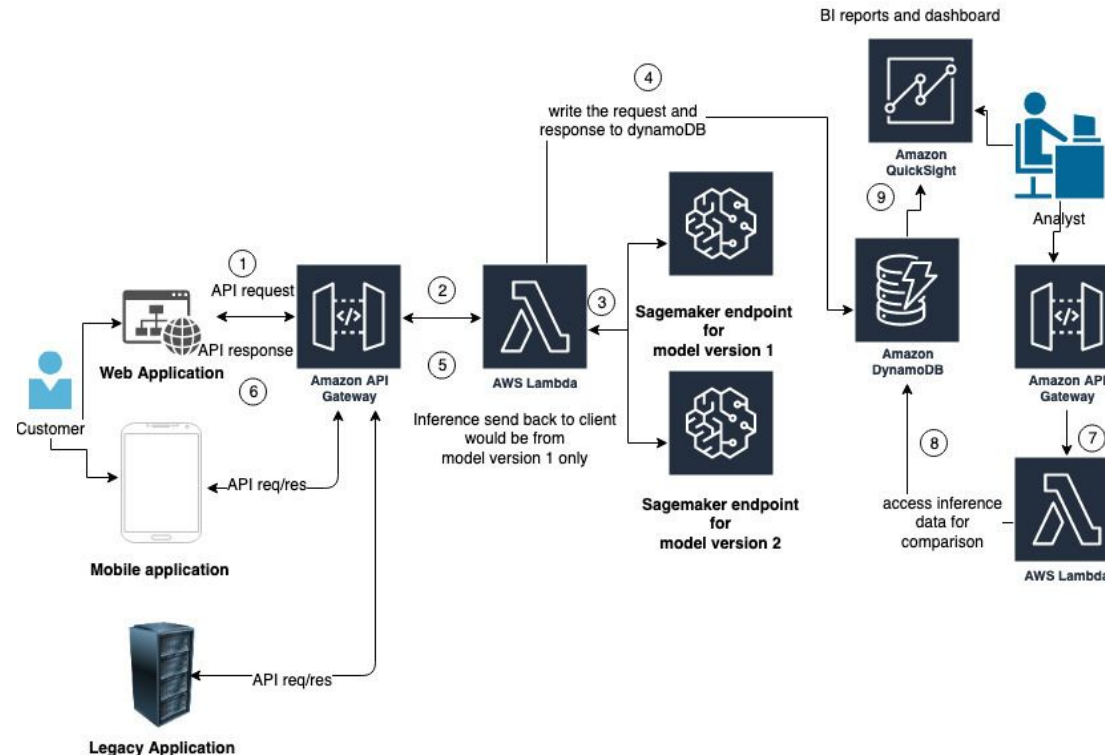
# A/B Testing



# Shadow deployment

Input data is run through both versions.

The older version is used for servicing the production, and the new version is used for testing and analysis.

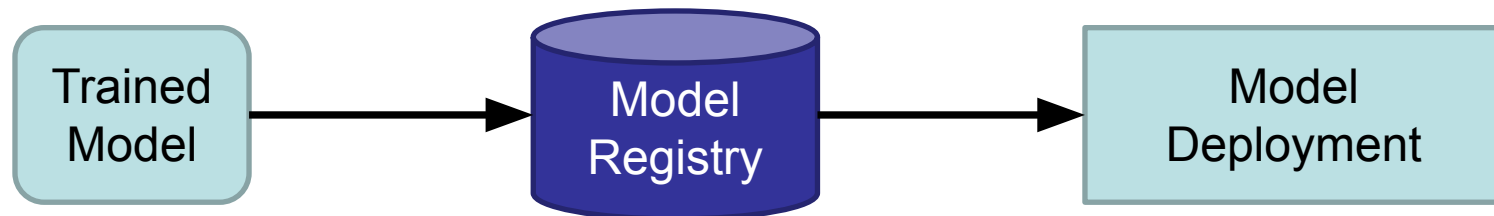


Source: <https://aws.amazon.com/blogs/machine-learning/deploy-shadow-ml-models-in-amazon-sagemaker/>

# Model Registry

# Model Deployment

- Not all trained models go into production environment.
  - We choose the best models from our model development
- We use Model Registry to manage the models in our production system.
- Model Registry is a metadata store for your machine learning models.
  - Catalog models for production
  - Manage model version
  - Manage approval status of a model



# Model Artefacts

# Model Artefacts

- The trained model and all other pre and post processing objects required for transformations
- `model_artefact_123`:
  - `trained_model`
  - `pre_processing_transformers`
  - `post_processing_transformers`
  - `calibrator_model`
  - `containers_specs / deployment_configs`
  - `secrets / endpoints configs`
  - `model_diagnostics / reports`
  - ...

# Lab

# Homework

Next week

# Quiz

Lock down browser required

# Project Proposal Presentations

Next week

# Lab Activities

1. Build a .py script that takes a snapshot date, loads model artefacts, make an inference and store the predictions into storage

Dataset to be provided by instructor

original source datasets for instructors: <https://www.kaggle.com/datasets/parisrohan/credit-score-classification>

An aerial night photograph of Singapore. The foreground shows the modern architecture of the Singapore Management University (SMU) campus, with its distinctive tiered, green-roofed buildings and glass facades. The middle ground is filled with the dense urban landscape of the city, featuring numerous skyscrapers and residential buildings, all illuminated with warm lights. The background shows the dark, silhouetted hills of the city. The overall scene is a vibrant display of urban development and architecture.

**Questions?**

**Thank You**